Quantitative chemistry

Count & conserve

1 Mass & formulae

Atoms are rearranged, not created/destroyed.Closed system: reactant mass = product mass.

Gas entering/leaving explains apparent mass change. M_r = sum of A_r values in formula; no unit.

Balance coefficients, never change formula subscripts.

2 Moles are a counting unit

One mole contains **6.02×10^{23}** stated particles. Molar mass in g/mol is numerically equal to M_r.

n = mass / molar mass. Particles = n × Avogadro constant.

Use the equation

3 Ratios & limiting reactants

mass | moles | ratio

Convert → compare → convert back

Mass → moles → equation ratio → required moles → mass.

Limiting reactant runs out and limits product; excess remains. Compare moles divided by each coefficient.

From masses: convert to moles, divide by smallest, obtain whole-number ratio.

4 Solutions & gases

1 dm^3 = 1,000 cm^3. Mass concentration = g / dm^3; molar concentration = mol / dm^3.

Titration: known cV → ratio → unknown n/V.

At RTP, gas volume = **n × 24 dm^3;** gas ratios follow equation coefficients.

Judge the outcome

5 Yield & atom economy

Yield = actual / theoretical × 100%. Losses, side reactions and incomplete reactions lower yield.

Atom economy = desired-product mass / total reactant mass from equation × 100%; include coefficients.

Compare rate, yield, waste, equilibrium and useful by-products.

6 Reliable measurements

Repeat, calculate mean and show spread. Half-range can estimate uncertainty when requested.

Percentage uncertainty = absolute uncertainty / reading × 100. Repeats reduce random effects, not systematic errors.

Keep units consistent and round only the final answer.